MILITARY SPECIFICATION

TARGET POSITION SIMULATOR MARK 26 MOD 0
(For Fire Control Systems Mk 112)

This specification has been approved by the
Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of the Target Position Simulator Mk 26 Mod 0. The Target Position Simulator is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 Unless otherwise specified, the following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shockproof Equipment, Class HI (High Impact) Shipboard Application, Tests for
MIL-A-5092	Adhesive Rubber (Synthetic and Reclaimed Rubber Base)
MIL-G-12803	Gasket Material, Non-metallic
MIL-C-16173	Corrosion Preventive Compound Solvent-Cutback, Cold Application

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MIL-E-17555	Preparation for Delivery of Electronic, Electrical, and Electro-Mechanical Equipment and Associated Repair Parts
MIL-F-18870(OS)	Fire Control Equipment, Naval Ship and Shore, General Specification for
MIL-I-26860	Indicator Humidity, Plug, Color Change

STANDARDS

MILITARY

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations for Shipboard Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment

DRAWINGS

NAVY DEPARTMENT

BUWEPS IL-2854925	Target Position Simulator Mk 26 Mod 0
BUWEPS DWG 2019166	Step Transmitter

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OR-11	Development of Packaging, Handling, Storage, and Transportations Systems for Weapons
WR-43	Preparation of Quality Assurance Provisions (Including Classification of Characteristics).

(Copies of specifications, standards, and drawings requested by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other Publications. The following documents form a part of the specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

Handbook H28 Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606. Orders for the above publication should cite the latest edition and supplements thereto.)

AMERICAN TRUCKING ASSOCIATION, INC. PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th Street N.W., Washington, D. C. 20036)

3. REQUIREMENTS

3.1 Description. The Target Position Simulator Mk 26 Mod 0 (hereinafter called the simulator) is installed in proximity to the fire control systems aboard submarines. The simulator supplies target position and own ship data to the various components of the fire control systems (computers, plotters, indicators) for on-board operational training. Target and own ship dial groupings, counters, and handcrank controls are similar and compatible in operation to those of submarine fire control computers.

Output signals are available by means of connectors at the rear of the enclosure; standard NAVORD synchros supply own course and speed, and target range, range rate, and bearing signals; own ship motion north and east are supplied in step transmitter form; "Mark" signals are supplied as 115V, 400 Hertz signals.

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3.2 General requirements.

3.2.1 First Article samples. Unless otherwise specified in the contract or order, First Article samples of the simulator shall be manufactured using the methods and procedures proposed for the production lot. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's methods are capable of yielding items that comply with the technical requirements of the contract.

3.2.2 Materials. Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of the simulator shall conform strictly to the detailed requirements indicated on the applicable drawings, specifications and publications listed in NAVORD IL-2854925.

3.2.3 Finishes, coatings, construction and dimensions. Finishes, coatings, construction and dimensions for the simulator shall be strictly in accordance with the applicable drawings, specifications and publications listed in NAVORD IL-2854925.

3.2.4 Moisture, fungus and corrosion. The simulator shall be rendered moisture, fungus and corrosion resistant in accordance with detail requirements specified in the drawings and specifications listed in NAVORD IL-2854925.

3.2.5 Marking. Marking of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawings listed in NAVORD IL-2854925.

3.2.6 Interchangeability. Individual parts of assemblies, fabricated in agreement with their applicable drawings and specifications shall be interchangeable with mating parts of the unit and other like units. If deviation greater or smaller than the tolerances shown on the drawings or specifications is required for interchangeability, or if hand fitting beyond that shown on applicable drawings is required, then changes in dimensions are necessary and revision and correction of the applicable drawings are required. (See 3.2.9.)

3.2.7 Requests for deviation. Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Naval Ordnance Systems Command or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation on the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the

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extent of the departure, the identifying number of the item, the contract (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.2.8 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.2.9 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.3 Performance requirements and product characteristics. The simulator shall meet the following performance requirements and product characteristics:

3.3.1 Power requirements. The simulator shall meet all of the performance requirements and product characteristics specified herein when energized with a 115 volt, 400 Hertz, Type I power supply as defined in MIL-F-18870.

3.3.2 Simulator output signals.

3.3.2.1 Generated range transmission. The simulator shall be capable of delivering generated range signals from synchro transmitters, type 23CX4 or equivalent, at 1,296,000 and 72,000 yards per turn at 90 volts, 400 Hertz, and a 23TDX4 or equivalent, at 2000 yards per turn at 90 volts, 400 Hertz.

3.3.2.2 Range mark transmission. The simulator shall be capable of delivering range mark signals at 115 volts, 400 Hertz.

3.3.2.3 Range rate transmission. The simulator shall be capable of delivering range rate signals from synchro transmitters, type 15CX4 or equivalent, at 360 and 10 knots per turn at 90 volts, 400 Hertz.

3.3.2.4 Range rate mark transmission. The simulator shall be capable of delivering range rate mark signals at 115 volts, 400 Hertz.

3.3.2.5 Generated relative target bearing transmission. The simulator shall be capable of delivering generated relative target bearing signals from a synchro transmitter type 23CX4 or equivalent at 360 degrees per turn, at 90 volts, 400 Hertz and a synchro transmitter, type 23TDX4 or equivalent, at 10 degrees per turn at 90 volts, 400 Hertz.

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3.3.2.6 Bearing mark transmission. The simulator shall be capable of delivering bearing mark signals at 115 volts, 400 Hertz.

3.3.2.7 Own ship course transmission. The simulator shall be capable of delivering own ship course signals from synchro transmitters type 23CX4 or equivalent at 360 and 10 degrees per turn at 90 volts, 400 Hertz.

3.3.2.8 Own ship speed transmission. The simulator shall be capable of delivering own ship speed signals from synchro transmitters type 23CX4 or equivalent at 100 and 10 knots per turn at 90 volts, 400 Hertz.

3.3.2.9 Own ship motion transmission. The simulator shall be capable of delivering own ship motion north and own ship motion east signals from step transmitters, NAVORD Dwg. 2019166 or equivalent, at 750 turns per nautical mile at 115 volts DC.

3.3.3 Simulator display and control.

3.3.3.1 Range. The simulator shall be capable of displaying ranges up to 72,000 yards with a resolution of 10 yards. A handcrank shall be used to enter initial values of range into the simulator.

3.3.3.2 Relative target bearing. The simulator shall be capable of displaying relative target bearing up to 360 degrees with a resolution of 0.1 degree. A handcrank shall be used to enter initial values of target bearing into the simulator.

3.3.3.3 Target course. The simulator shall be capable of displaying target course up to 360 degrees with a resolution of 0.1 degree. A handcrank shall be used to enter values of target course into the simulator.

3.3.3.4 Own ship course. The simulator shall be capable of displaying own ship course up to 360 degrees with a resolution of 0.1 degree. A handcrank shall be used to enter values of own ship course into the simulator.

3.3.3.5 Own ship speed. The simulator shall be capable of displaying own ship speed up to 50 knots with a resolution of 0.1 knot. A handcrank shall be used to enter values of own ship speed into the simulator.

3.3.3.6 Target speed. The simulator shall be capable of displaying target speed up to 50 knots with a resolution of 0.1 knots. A handcrank shall be used to enter values of target speed into the simulator.

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3.3.3.7 Range rate. The simulator shall be capable of displaying opening and closing range rates up to 100 knots with a resolution of 0.2 knots.

3.3.3.8 Mark lamps. Range, range rate, and bearing mark panel indicator lamps shall light when their respective switches are operated.

3.3.3.9 Error control. Range and bearing error switches shall control the excitation of the range and bearing error generators.

3.3.3.10 Power-on lamp. A power-on indicator shall light when power is applied in any simulator operating mode except off.

3.3.3.11 Power-mode selection. A four-position mode selector switch shall provide off, standby, static, and dynamic modes of operation. In the off mode of operation, power shall be disconnected from all simulator output connections; in the standby mode, warm-up power shall be applied; in the static mode, output data shall be controlled manually by handcrank; in the dynamic mode, target position data, after initial conditions are inserted into the simulator by means of handcranks, shall be generated by the simulator as functions of time.

3.3.3.12 Illumination. All panel illumination lamps shall light under control of a panel illumination switch. Illumination shall meet the dark adaptation requirements of MIL-F-18870(OS).

3.3.4 Simulator computations.

3.3.4.1 Range and bearing generation. The simulator shall be capable of computing range and bearing signals within the specified errors of 4.4.3.9.

3.3.4.2 Own ship motion generation. The simulator shall be capable of computing the components of own ship motion north and own ship motion east with a total vector accuracy of 10 yards per minute.

3.3.4.3 Range rate generation. The simulator shall be capable of computing range rate with an accuracy of plus or minus 0.3 knots.

3.3.5 Error generation.

3.3.5.1 Range error generation. The simulator shall be capable of delivering range signals which are modulated to produce an error envelope up to plus and minus 125 yards.

3.3.5.2 Bearing error generation. The simulator shall be capable of delivering bearing signals which are modulated to produce an error envelope up to plus and minus 2.5 degrees.

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3.3.6 Dielectric. The simulator shall be capable of withstanding the dielectric tests specified in 4.4.3.2.

3.3.7 Insulation resistance. The simulator shall be capable of withstanding the insulation resistance tests specified in 4.4.3.3.

3.4 Environmental requirements. The simulator shall be capable of meeting the following environmental requirements:

3.4.1 Vibration.

3.4.1.1 Vibration, transportation. The simulator, packaged and packed as defined in Section 5, shall be capable of withstanding the vibration tests specified in Ordnance Requirement OR-11 and thereafter shall show no damage and shall meet all the requirements of 3.3.

3.4.1.2 Vibration, operating condition. The simulator shall withstand the Type I vibration requirements specified in MIL-STD-167, shall function during and after vibration tests, and thereafter shall show no damage and shall meet all the requirements of 3.3.

3.4.2 Shock.

3.4.2.1 Shock, transportation. The simulator, packaged and packed as defined in Section 5, shall be capable of withstanding the appropriate rough handling tests specified in Ordnance Requirement OR-11, and thereafter shall show no damage and shall meet all the requirements of 3.3.

3.4.2.2 Shock, energized condition. The simulator shall be capable of being mounted on a standard test fixture and subjected to the type A shock tests for grade B, Class I, lightweight, hull-mounted equipment, as described and defined in MIL-S-901C. The simulator will have passed the test when all of the acceptability criteria as defined in MIL-S-901C have been satisfied.

3.4.3 Temperature.

3.4.3.1 Temperature, operating condition. The simulator shall be capable of being subjected to the temperature cycling specified in MIL-F-18870 for Class 4 equipment for ambient temperatures between plus 20 degrees C. and 50 degrees C., and shall show no damage or corrosion and shall meet all the requirements of 3.3.

3.4.3.2 Temperature, non-operating condition. The simulator shall be capable of being subjected to the temperature cycle specified in MIL-F-18870, Non-operating Condition, and thereafter shall show no damage or corrosion and shall meet all the requirements of 3.3.

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3.4.4 Humidity. The simulator shall be capable of being operated at relative humidities up to 95%, and being tested in accordance with the procedures of MIL-F-18870, and thereafter shall show no damage and shall meet all the requirements of 3.3.

3.4.5 Inclination. The simulator shall be capable of being subjected to the inclination tests of MIL-F-18870, and thereafter shall show no damage and shall meet all the requirements of 3.3.

3.4.6 Degree of enclosure. The simulator shall be capable of withstanding the fogproof tests specified in MIL-STD-108.

3.4.7 Air transportation. The simulator shall be capable of being shipped by means of air transportation at altitudes up to 30,000 feet in non-pressurized compartments and being tested for this ability in accordance with the procedures of Altitude Test (transport) of MIL-F-18870; thereafter, the simulator shall show no damage and meet all the requirements of 3.3.

3.5 Workmanship. Workmanship shall be in accordance with MIL-STD-454 requirement 9.

3.5.1 The standards of workmanship exhibited in any approved First Article sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 First Article sampling. A First Article sample of 1 simulator manufactured in accordance with 3.2 shall be subjected to the First Article tests detailed in 4.3 at an activity designated by the procuring activity. Acceptance of the First Article sample shall be based on no defects in the sample. Further production of the simulator by the supplier, prior to approval of the First Article sample shall be at the supplier's risk.

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4.2.2 Periodic production sampling. Periodic production samples shall be selected from units that have satisfactorily passed the acceptance inspections of 4.4. These samples shall be subjected to the tests of 4.3 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production, randomly select 1 unit from each group of 10 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2.

Step 2. Randomly select 1 unit from each group of 25 units. Upon satisfactory compliance of samples from 3 consecutive groups, shift to Step 3.

Step 3. Randomly select 1 unit from each group of 50 units. Continue with Step 3 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(a) Determine the cause of the defect.

(b) Take the necessary actions to correct or eliminate the defect from all units on hand and, of so directed by the procuring activity, from those units previously shipped.

(c) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.3 First Article and Periodic Production Tests. Samples selected in accordance with 4.2.1 and 4.2.2 shall be subjected to the tests of Table I and shall meet the applicable requirements of 3.3 and 3.4.

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TABLE I

FIRST ARTICLE AND PERIODIC PRODUCTION TESTS

TEST	REQUIREMENT REFERENCE	TEST REFERENCE	CLASSIFICATION OF CHARACTERISTICS
Power Source Variation	3.3.1	MIL-F-18870	M108
Vibration, Transportation	3.4.1.1	OR-11	M109
Vibration, Operating	3.4.1.2	MIL-STD-167	M110
Shock, Transportation	3.4.2.1	OR-11	M111
Shock, Operating	3.4.2.2	MIL-S-901	C3
Temperature, Operating	3.4.3.1	MIL-F-18870	M112
Temperature, Storage	3.4.3.2	MIL-F-18870	M113
Humidity	3.4.4	MIL-F-18870	M114
Inclination	3.4.5	MIL-F-18870	M115
Degree of Enclosure	3.4.6	MIL-STD-108	M116
Altitude, Transportation	3.4.7	MIL-F-18870	M117

4.4 Quality Conformance inspection. Each simulator delivered under contract shall be subjected to the tests and examinations listed in Table II. Failure to satisfactorily pass any test or examination specified herein shall be considered cause for rejection.

4.4.1 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43A. Tests and examinations that verify critical characteristics are identified by the symbol (C), and major characteristics, by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified Minor.

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TABLE II

QUALITY CONFORMANCE TESTS

EXAMINATION AND TEST	REQUIREMENT REFERENCE	TEST REFERENCE	CLASSIFICATION OF CHARACTERISTICS
General Examination	3.5	4.4.3.1	--
Dielectric Strength	3.3.6	4.4.3.2	C1
Insulation Resistance	3.3.7	4.4.3.3	C2
Synchro Transmission System	3.3.2	4.4.3.4	M101
Error Transmission System	3.3.5	4.4.3.5	M102
Own Ship Motion Accuracy	3.3.4.2	4.4.3.6	M103
Range Rate Accuracy	3.3.4.3	4.4.3.7	M104
Range and Bearing Alignment	3.3.4.1	4.4.3.8	M105
Range and Bearing Accuracy	3.3.4.1	4.4.3.9	M106
Mark Output Circuit	3.3.2	4.4.3.10	M107

4.4.2 Test conditions. Unless otherwise specified, the simulator shall be subjected to quality conformance inspection under the following conditions:

4.4.2.1 Temperature. Room ambient 65 degrees F. to 95 degrees F.

4.4.2.2 Altitude. Normal ground.

4.4.2.3 Vibration. None

4.4.2.4 Humidity. Room ambient to 95 percent relative (maximum).

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4.4.2.5 Test equipment. Items of test equipment required to perform quality conformance testing or their equivalents are listed in Table III.

TABLE III

TEST EQUIPMENT

Test Equipment	Identification
Multimeter	AN/PSM-4
High Potential Tester	Ideal Industries N041-001
Meg ohmmeter	Freed Type 1020-B
VTVM	Ballantine Model 300H
Timer (9999.9 seconds)	Cramer Type 635
Recorder	Sanborn 67-1200

4.4.2.6 Test Circuits. Three test circuits, Figures 1, 2 and 3 of this specification, are used in the performance of quality conformance testing as follows:

(a) Figure 1, Transmission Accuracy and Continuity Test Circuit #1, is used for tests of 4.4.3.4 and 4.4.3.5.

(b) Figure 2, Own Ship Motion Test Circuit #2, is used for tests of 3.3.4.2.

(c) Figure 3, Mark Output Test Circuit #3, is used for tests of 3.3.2.

4.4.3 Quality conformance test procedures.

4.4.3.1 General examination.

(a) Power-mode selector switch check. (see 3.3.3.11)

(b) Power-on lamp check. (see 3.3.3.10)

(c) Illumination switch check. (see 3.3.3.12)

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(d) Mark switch and mark lamp checks. (see 3.3.3.8)

(e) Correct fuse valve checks. (see 3.2.2)

(f) Workmanship. (see 3.5)

(g) Fungus and moisture treatment. (see 3.2.4)

4.4.3.2 Dielectric strength (C1). The dielectric test will consist of the application of 60 Hertz, 1000 volt RMS power, in accordance with MIL-F-18870, and with the removal or disconnection of the following components:

(a) Remove subassemblies A4A1, A4A2, A4A3, A4A4, and A4A5.

(b) Disconnect all synchro transmitter rotors.

(c) Disconnect all servo motors.

The high potential test leads shall be connected between the wires of power input connector A1J1 and ground with terminals A, B and C of the connector shorted together; POWER-MODE switch in DYNAMIC position.

4.4.3.3 Insulation resistance (C2). Upon completion of the dielectric strength test of 4.4.3.2, the insulation resistance at the power input plug A1J1 shall be measured. This resistance shall be a minimum of 3 megohms.

4.4.3.4 Synchro transmission system checks (M101). The alignment of the synchro transmitters shall conform to the electrical zero settings listed in Table IV, and shall be checked by means of an AC voltmeter connected into the output signal connectors A1J2 and A1J3 at terminals corresponding to terminals S1 and S3 of the synchro as shown in Test Circuit #1, Figure 1. Gearing backlash shall be checked by approaching the voltmeter nulls from both the cw and ccw directions by means of the handcranks and the backlash shall be within the limits shown in Table IV. Continuity between connectors A1J2 and A1J3 and each of the synchros (terminals S1-S2 and S2-S3) shall be checked by means of AC voltmeter measurements of approximately 78 volts at the continuity check terminals shown in Table IV. All measurements shall be made with the POWER-MODE switch in the STATIC position, and with the RANGE ERROR and BEARING ERROR switches in the OFF position.

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TABLE IV

TRANSMISSION SYSTEM CHECKS

SYNCHRO FUNCTION	ELECTRICAL ZERO SETTING	BACKLASH LIMITS	ELECTRICAL ZERO TERMINALS	CONTINUITY CHECK TERMINALS
OWN SHIP SPEED	0.0 knots			
10 kt/turn		±0.1 kt	A1J3-D & F	A1J3-D & E A1J3-E & F
100 kt/turn		±0.2 kt	A1J3-A & C	A1J3-A & B A1J3-B & C
OWN SHIP COURSE	000.0 degrees			
10°/turn		±0.1°	A1J3-K & M	A1J3-K & L A1J3-L & M
360°/turn		±0.2°	A1J3-G & J	A1J3-G & H A1J3-H & j
RELATIVE BEARING	000.0 degrees			
10°/turn		±0.1°	A1J2-D & F	A1J2-D & E A1J2-E & F
360°/turn		±0.2°	A1J2-A & C	A1J2-A & B A1J2-B & C
RANGE	10,000 yards			
2,000 yd/turn		±10 yds	A1J2-N & Q	A1J2-N & P A1J2-P & Q
72,000 yd/turn		±50 yds	A1J2-K & M	A1J2-K & L A1J2-L & M
1,296,000 yd/turn		±750 yds	A1J2-G & J	A1J2-G & H A1J2-H & J
RANGE RATE	0.0 knots			
10 kts/turn		±0.1 kt	A1J2-U & W	A1J2-U & V A1J2-V & W
360 kts/turn		±0.2 kt	A1J2-R & T	A1J2-R & S A1J2-S & T

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4.4.3.5 Error transmission system checks (M102). The Range and Bearing error transmission systems shall be checked, using Test Circuit #1, Figure 1, as follows:

(a) Set controls as follows:

OWN SPEED	00.0 knots
TARGET SPEED	00.0 knots
BEARING	00.0 degrees
RANGE	10,000 yards
OWN COURSE	00.0 degrees
TARGET COURSE	00.0 degrees
POWER-MODE Switch	DYNAMIC

(b) Connect AC voltmeter across pins D and F of connector A1J2.

(c) Turn BEAR ERROR switch to the ON position and note the amplitude of the two voltage maxima which occur over a three-minute period. The two maxima shall be equal to each other within ±1 volt and have a voltage amplitude between 85 and 90 volts.

(d) Connect AC voltmeter across pins N and Q of connector A1J2.

(e) Turn RANGE ERROR switch to the ON position and note the amplitude of the two voltage maxima which occur over a three-minute period. The two maxima shall be equal to each other within ±2 volts and have a voltage amplitude between 30 and 35 volts.

4.4.3.6 Own ship motion accuracy (M103). The North and East stepper switches produce 115VDC pulses proportional to the rate of own ship travel. In the below tests, either a paper chart recorder or preset counter may be used to count the number of pulses per minute. Test circuit #2, Figure 2, is used in the performance of this test.

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TEST PROBLEM NO. 1

(a) Set controls as follows:

OWN COURSE	45°
OWN SPEED	30 knots
POWER-MODE Switch	DYNAMIC

(b) Record or count pulses over a one-minute period.

(c) The number of pulses per minute both North and East should be 265-1/8. Subtract the observed number of pulses North and number of pulses East from this figure and obtain the vector sum of the error. The permissible vector error is 3-3/4 pulses per minute.

TEST PROBLEM NO. 2

(a) Set controls as follows:

OWN COURSE	089.8°
OWN SPEED	50 knots

(b) Observe that the North channel electronic integrator is operating at low signal levels as evidenced by an occasional pulse on the recorder or counter.

(c) Change OWN COURSE setting to 000.2°.

(d) Observe that the East channel electronic integrator is operating at low signal levels as evidenced by an occasional pulse on the recorder or counter.

4.4.3.7 Range rate accuracy (M104). Test problems are used to check the range rate servo positioning accuracy and acceleration. The problems are set-up in the STATIC mode and started by switch to the DYNAMIC mode.

TEST PROBLEM NO. 1

(a) Set controls in the STATIC mode as follows:

OWN COURSE	090°
OWN SPEED	25 knots
TARGET COURSE	090°
TARGET SPEED	25 knots
RELATIVE BEARING	000°
RANGE	1,000 yards

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(b) In the DYNAMIC mode, the range rate counter shall read 00.0 knots plus or minus 0.3 knots.

(c) Switch to STATIC mode and set controls as follows:

RELATIVE BEARING	180°
OWN COURSE	270°

(d) In the DYNAMIC mode, the range rate counter shall slew to indicate 50 knots plus or minus 0.3 knots OPENING in less than 2.5 seconds.

TEST PROBLEM NO. 2

(a) Set controls on the STATIC mode as follows:

OWN COURSE	000°
OWN SPEED	10 knots
TARGET COURSE	180°
TARGET SPEED	20 knots
RELATIVE BEARING	000°
RANGE	30,000 yards

(b) In the DYNAMIC mode, the range rate counter shall slew to indicate 30 knots plus or minus 0.3 knots CLOSING. Slowly decrease TARGET SPEED to indicate 00.0 knots, and observe that the counter follows smoothly and comes to rest at 10 knots plus or minus 0.3 knots.

4.4.3.8 Range and bearing alignment (M105). Overall alignment and stability of the range and bearing servos shall be tested as follows:

(a) Set controls in the STATIC mode as follows:

OWN COURSE	225°
OWN SPEED	10 knots
TARGET COURSE	225°
TARGET SPEED	10 knots
RANGE	2,000 yards

(b) Switch to DYNAMIC mode.

(c) Observe that the range and bearing dials remain within 0.25 degrees and 25 yards of the above settings over a period of 5 minutes.

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(d) Slowly vary the TARGET BEARING handcrank, stopping at each 45° increment of bearing. There shall be no perceptible motion of the bearing and range dials at any setting of the handcrank.

4.4.3.9 Range and bearing accuracy (M106). Eight test problems are used to check range and bearing servo accuracies. Set-up of controls for each test problem is indicated in Table V. Tables VI through XIII are data sheets, corresponding to each of the test problems, which indicate dial readings during problem generation and the allowed errors. All problems are set up in an identical manner; the range is allowed to generate until it reaches the range indicated at time zero on the data sheets while the TARGET BEARING handcrank is used to maintain the initial relative bearing setting until time zero is reached. At time zero, the handcrank is released and a stopwatch or timer is started. Data observed on the range and bearing dials is checked against the problem data sheet at each one-minute increment of time. No individual error shall exceed 300% of the allowed error and the average error for the problem shall not exceed 100% of the allowed average error.

TABLE V

Problem Number	Own Course Degrees	Own Speed Knots	Target Speed Knots	Target Course Degrees	Zero Time Range Yards	Starting Range Yards	Initial Relative Bearing Degrees
1	210.0	2	5	35.0	2,500	2,800	20.0
2	000.0	10	15	235.0	8,000	8,300	45.0
3	240.0	7	20	340.0	8,000	8,300	310.0
4	60.0	10	20	355.0	6,000	6,300	75.0
5	30.0	15	30	15.0	7,000	7,300	140.0
6	270.0	5	25	90.0	24,000	24,000	5.0
7	90.0	20	40	270.0	60,000	60,300	20.0
8	200.0	10	30	300.0	60,000	60,300	310.0

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TABLE VI

R & BR, TEST PROBLEM NUMBER 1

	RELATIVE BEARING				RANGE			
Time Min.	Calculated Degrees	Observed Degrees	Error Degrees	Allowed Error Degrees	Calculated Yards	Observed Yards	Error Yards	Allowed Error Yards
0	20.00			--	2500			--
1	21.68			.3	2275			16
2	23.73			.4	2051			22
3	26.28			.5	1832			28
4	29.51			.5	1616			35
5	33.72			.6	1408			41
6	39.37			.7	1209			47
7	47.12			.9	1026			52
8	57.88			1.0	871			58
9	72.57			1.2	757			64
10	90.72			1.5	708			69
11	109.45			1.7	735			74
12	125.30			1.9	831			78
13	137.18			2.1	977			83
14	145.72			2.2	1152			87
15	151.90			2.3	1347			91

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TABLE VII

R & BR, TEST PROBLEM NUMBER 2

RELATIVE BEARING					RANGE			
Time Min.	Calculated Degrees	Observed Degrees	Error Degrees	Allowed Error Degrees	Calculated Yards	Observed Yards	Error Yards	Allowed Error Yards
0	45.00			--	8000			--
1	46.19			.3	7264			19
2	47.65			.4	6532			27
3	49.47			.4	5806			36
4	51.81			.5	5087			45
5	54.91			.6	4379			53
6	59.22			.7	3686			62
7	65.43			.8	3025			70
8	74.92			.9	2420			78
9	89.97			1.1	1921			85
10	112.62			1.2	1632			92
11	139.02			1.7	1664			97
12	160.25			2.0	2002			100
13	174.07			2.2	2526			102
14	182.80			2.3	3149			104
15	188.58			2.4	3812			106

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TABLE VIII

R & BR, TEST PROBLEM NUMBER 3

	RELATIVE BEARING				RANGE			
Time Min.	Calculated Degrees	Observed Degrees	Error Degrees	Allowed Error Degrees	Calculated Yards	Observed Yards	Error Yards	Allowed Error Yards
0	310.00			--	8000			--
1	311.24			.3	7265			19
2	312.75			.4	6534			27
3	314.64			.4	5809			36
4	317.06			.5	5092			45
5	320.28			.6	4388			53
6	324.72			.7	3699			62
7	331.10			.8	3045			70
8	340.78			1.0	2449			78
9	355.90			1.2	1963			85
10	18.12			3.7	1688			92
11	43.58			3.5	1727			96
12	64.29			3.3	2061			100
13	77.83			3.2	2579			102
14	86.60			3.2	3192			104
15	92.43			3.2	3855			106

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TABLE IX

R & BR, TEST PROBLEM NUMBER 4

RELATIVE BEARING					RANGE			
Time Min.	Calculated Degrees	Observed Degrees	Error Degrees	Allowed Error Degrees	Calculated Yards	Observed Yards	Error Yards	Allowed Error Yards
0	75.00			--	6000			--
1	73.85			.3	5397			18
2	72.42			.4	4796			26
3	70.58			.4	4199			34
4	68.13			.5	3608			42
5	64.73			.6	3026			50
6	59.72			.7	2457			58
7	51.82			.8	1920			65
8	38.33			1.0	1447			73
9	14.75			1.3	1119			80
10	342.37			3.4	1081			85
11	316.13			3.2	1356			88
12	300.82			3.1	1807			91
13	291.98			3.1	2334			93
14	286.47			3.1	2895			95
15	282.78			3.1	3475			98

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TABLE X

R & BR, TEST PROBLEM NUMBER 5

RELATIVE BEARING					RANGE			
Time Min.	Calculated Degrees	Observed Degrees	Error Degrees	Allowed Error Degrees	Calculated Yards	Observed Yards	Error Yards	Allowed Error Yards
0	140.00			--	7000			--
1	139.09			.3	6471			18
2	138.02			.4	5944			25
3	136.75			.4	5419			33
4	135.20			.5	4898			41
5	133.28			.6	4380			48
6	130.85			.6	3868			56
7	127.78			.7	3366			63
8	123.43			.8	2877			71
9	117.52			.9	2412			78
10	108.85			1.1	1983			85
11	95.97			1.2	1623			92
12	77.35			1.5	1386			98
13	54.57			1.8	1338			103
14	33.55			2.0	1497			108
15	18.20			2.2	1810			111

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TABLE XI

R & BR, TEST PROBLEM NUMBER 6

RELATIVE BEARING					RANGE			
Time Min.	Calculated Degrees	Observed Degrees	Error Degrees	Allowed Error Degrees	Calculated Yards	Observed Yards	Error Yards	Allowed Error Yards
0	5.00			--	24,000			--
1	5.22			.3	22,991			20
2	5.47			.4	21,982			30
3	5.72			.4	20,973			40
4	6.02			.5	19,965			50
5	6.33			.5	18,957			60
6	6.70			.6	17,951			70
7	7.08			.6	16,945			80
8	7.53			.7	15,939			90
9	8.05			.7	14,935			100
10	8.63			.8	13,934			110
11	9.32			.8	12,932			120
12	10.10			.9	11,933			130
13	11.03			1.0	10,937			140
14	12.15			1.0	9,945			150
15	13.50			1.1	8,956			160

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TABLE XII

R & BR, TEST PROBLEM NUMBER 7

RELATIVE BEARING					RANGE			
Time Min.	Calculated Degrees	Observed Degrees	Error Degrees	Allowed Error Degrees	Calculated Yards	Observed Yards	Error Yards	Allowed Error Yards
0	20.00			--	60,000			--
1	20.68			.3	58,099			25
2	21.42			.3	56,208			39
3	22.20			.4	54,326			53
4	23.03			.5	52,455			68
5	23.93			.5	50,596			82
6	24.90			.6	48,750			96
7	25.93			.7	46,919			111
8	27.07			.7	45,106			125
9	28.28			.8	43,310			139
10	29.60			.8	41,537			152
11	31.05			.9	39,787			166
12	32.62			1.0	38,066			180
13	34.35			1.0	36,374			193
14	36.23			1.1	34,720			206
15	38.30			1.2	33,107			220

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TABLE XIII

R & BR, TEST PROBLEM NUMBER 8

RELATIVE BEARING					RANGE			
Time Min.	Calculated Degrees	Observed Degrees	Error Degrees	Allowed Error Degrees	Calculated Yards	Observed Yards	Error Yards	Allowed Error Yards
0	310.00			--	60,000			--
1	310.23			.3	58,906			21
2	310.48			.3	57,813			31
3	310.75			.4	56,721			41
4	311.02			.5	55,630			52
5	311.30			.5	54,541			62
6	311.60			.6	53,452			73
7	311.90			.6	52,366			83
8	312.22			.7	51,280			94
9	312.55			.7	50,197			104
10	312.90			.8	49,115			115
11	313.25			.8	48,035			125
12	313.63			.9	46,957			135
13	314.03			.9	45,882			146
14	314.43			1.0	44,811			156
15	314.87			1.0	43,737			166

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4.4.3.10 Mark output circuit test (M107). Using Test Circuit #3 check range, range rate, and bearing output circuitry as follows:

(a) Range. Connect the voltmeter to pin X of connector A1J2 and operate the RANGE mark switch to ATF and MARK positions; the voltmeter shall indicate between 110 and 120 VAC in either position.

(b) Range rate. Connect the voltmeter to pin Y of connector A1J2 and operate the RANGE RATE mark switch to ATF and MARK positions; the voltmeter shall indicate between 110 and 120 VAC in either position.

(c) Bearing. Connect the voltmeter to pin Z of connector A1J2 and operate the BEARING mark switch to ATF and MARK positions; the voltmeter shall indicate between 110 and 120 VAC in either position.

4.5 Failure of the simulator to meet any of the requirements and tests of this specification shall be considered cause for rejection.

4.6 Waiver of Tests. The transportation tests specified in Table I may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.7 Functional Tests. At the conclusion of the transportation tests specified in Table I, functional and other tests, as applicable, shall be conducted on the simulator to determine freedom from operational malfunction and compliance with all the requirements of Section 3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 Level A. In addition to the following, Level A packaging shall be in accordance with MIL-E-17555.

5.1.1.1 Cleaning. The exterior surfaces only of the simulator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. The exterior surfaces only of the simulator shall be dried in accordance with procedure D-1 or D-4 of MIL-P-116.

5.1.1.3 Preservation application. All exterior bare ferrous metal surfaces shall be coated with preservation compound conforming to MIL-C-16173, Grade 2.

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5.1.1.4 Unit packaging. The simulator shall be individually packaged in accordance with Method IIa of MIL-P-116. Movable parts, susceptible to damage, shall be cushioned, secured, blocked and braced to prevent movement prior to packaging. Material for gaskets used to fabricate the package shall conform to MIL-G-12803, Type I, Identification No. P1141-A. The adhesive for cementing the gasket shall conform to MIL-A-5092, Type III. The humidity indicator for the package shall conform to MIL-I-26860, Type II.

5.1.1.5 Cushioning. The unit package and pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the simulator will meet all the requirements of Section 3 after being subjected to the transportation tests of Table I.

At the time of packaging or packing, the moisture content of the cushioning material shall not be more than 5 per cent. The pH range of the material shall be between 6.0 and 9.0

5.1.2 Level B. Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, drying and preservative application of the simulator shall be in accordance with 5.1.

5.1.3.2 Unit packaging. The simulator shall be individually packaged in accordance with the manufacturer's commercial practice, when such practices have been proven by testing to conform to all applicable requirements. The packaging shall be adequate to provide protection from physical damage, dust, dirt, corrosion and other forms of deterioration during direct shipment from supply source to the first receiving activity for immediate use. The method used shall be commensurate with shipping at the lowest practicable commercial rate.

5.2 Packing.

5.2.1 Level A. The simulator packaged as specified in paragraph 5.1, shall be individually packed in (overseas type) wood cleated plywood or nailed wooden box conforming to PPP-B-601, Style A or B or PPP-B-621, Class 2, Style 2 or 4, respectively. The simulator shall be cushioned, blocked and braced to provide protection from physical damage during overseas shipment, extended storage, multiple handling and domestic reshipment. The shipping container shall be provided with an inspection port for viewing the humidity indicator. The shipping containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.2 Level B. Not applicable.

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5.2.3 Level C. The simulator packaged as specified in 5.1, shall be individually packed in a manner to insure carrier acceptance for safe delivery at destination at the lowest rate. The shipping containers shall conform with the Uniform Freight Classification Rules, Motor Freight Classification Rules or the regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.1 Special marking. The following special marking shall appear on the top, one (1) side and one (1) end of each shipping container:

DELICATE INSTRUMENT
HANDLE WITH CARE

6. NOTES

6.1 Intended Use. The Target Position Simulator Mark 26 Mod 0 is intended for use in the Fire Control Systems Mark 112.

6.2 Ordering Data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) First Article - whether First Article sample is required (see 3.2.1).

(c) Required level of preservation and packaging.

(d) Required level of packing.

(e) Exceptions to this specification.

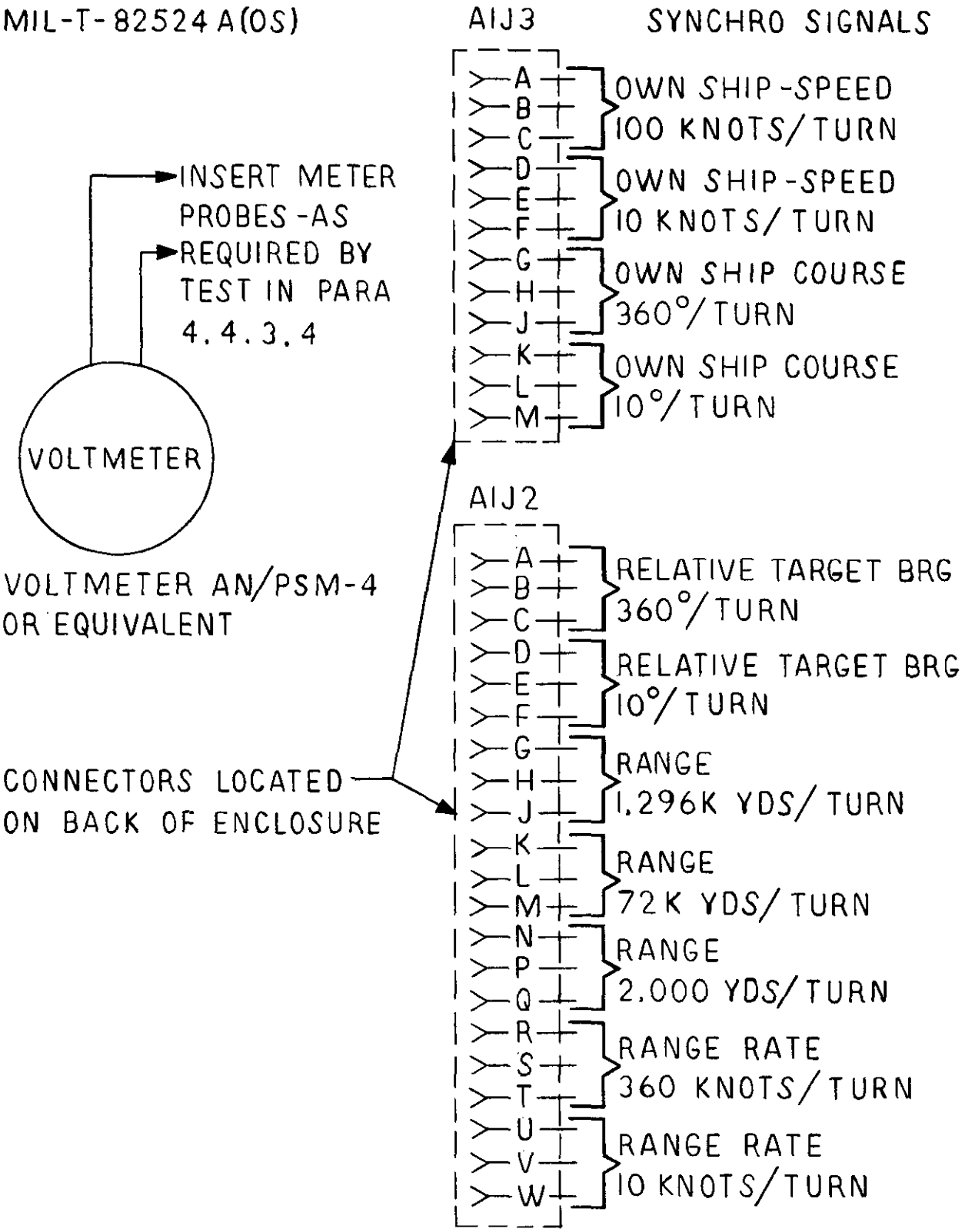
(f) Facility assigned responsibility for First Article inspection.

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and Options in MIL-P-116 and 4.6 of this specification.

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6.2.2 Guarantee. The contract or order shall include a requirement for a guarantee stating that any simulator which fails to meet the requirements of this specification within one year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

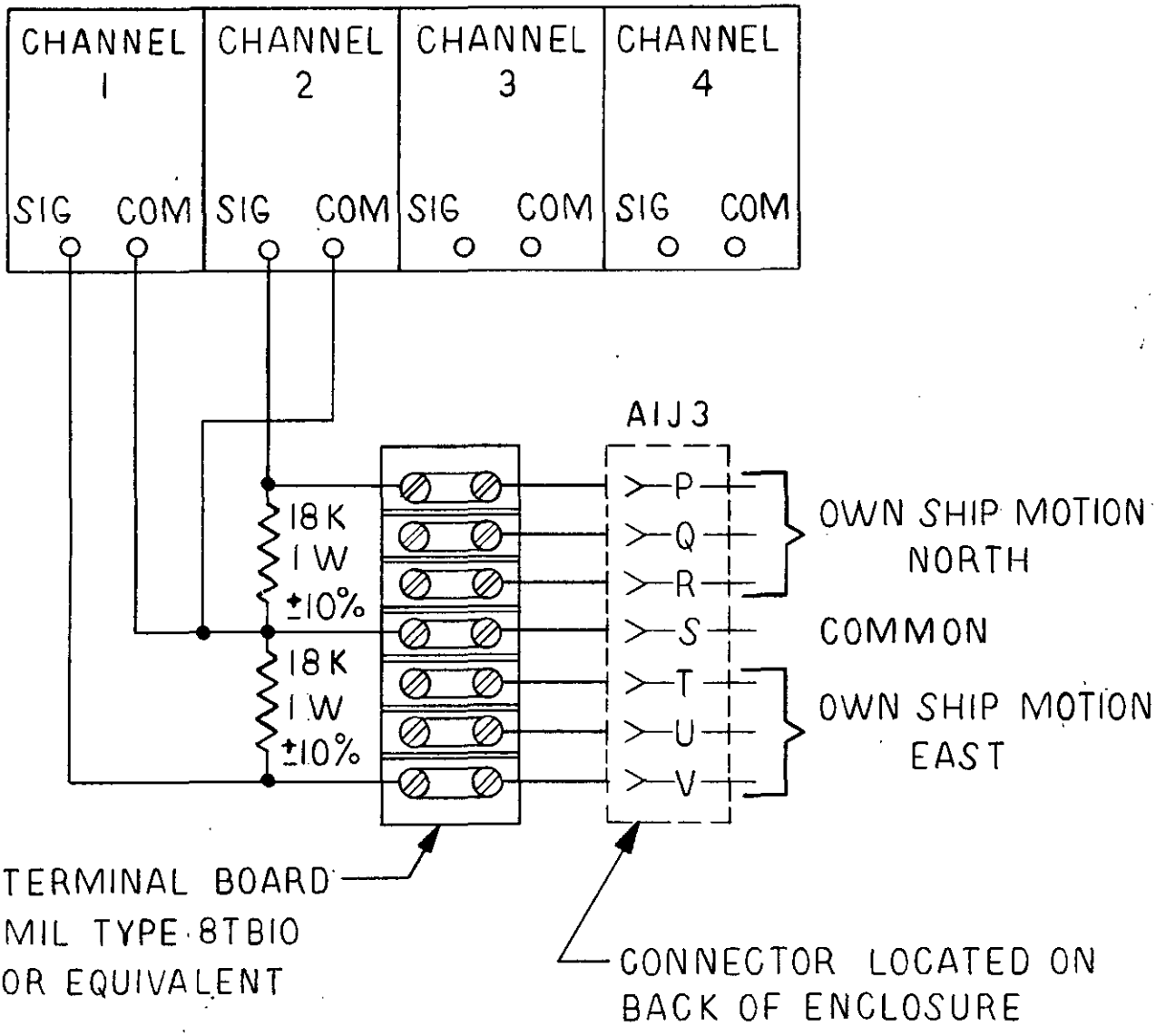
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TRANSMISSION ACCURACY & CONTINUITY
TEST CIRCUIT NO.1
FIGURE 1

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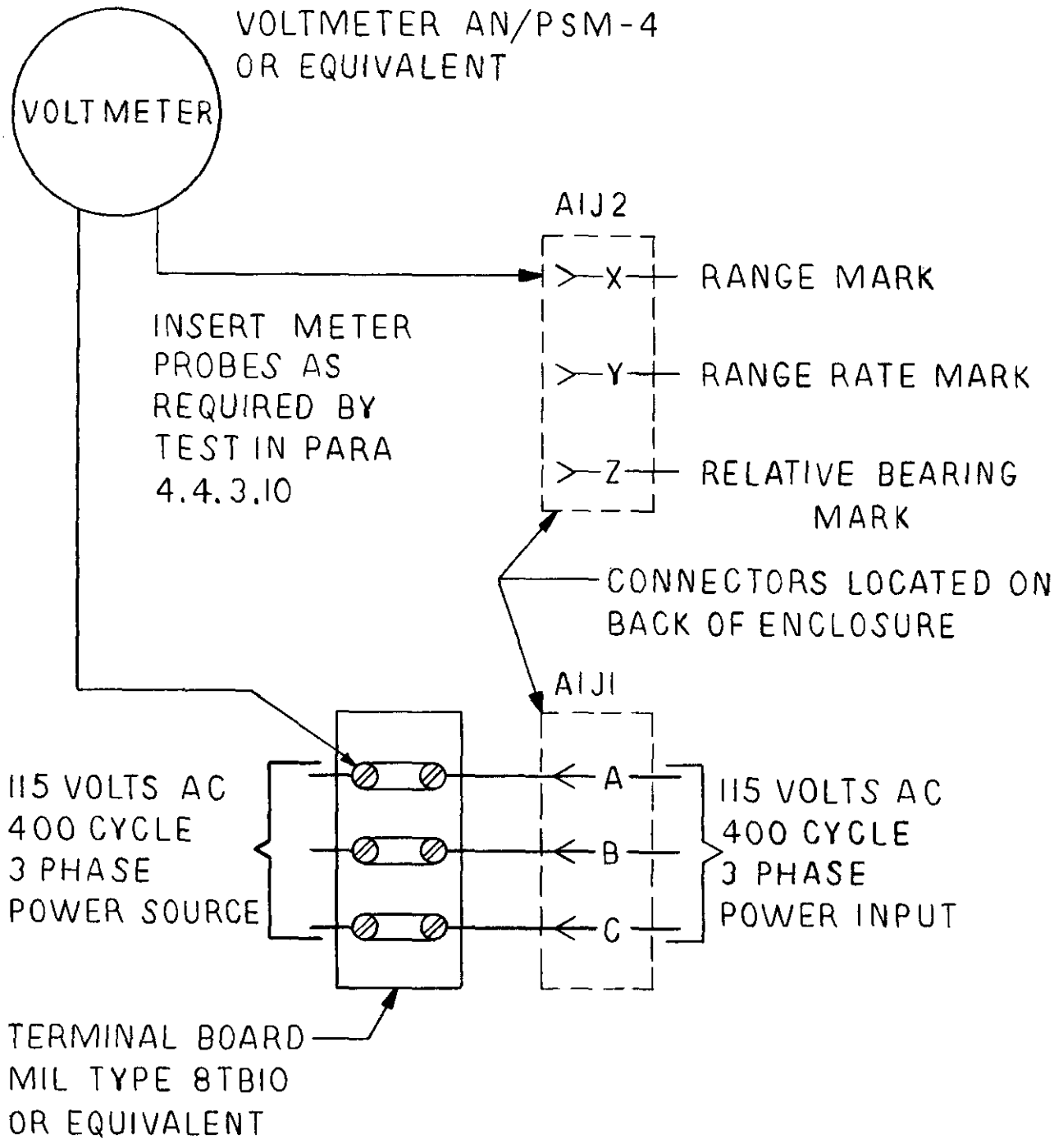
SANBORN MODEL 67-1200
FOUR CHANNEL RECORDER
OR EQUIVALENT



OWN SHIP MOTION
TEST CIRCUIT NO. 2

FIGURE 2

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MARK OUTPUT
TEST CIRCUIT NO. 3

FIGURE 3

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Naval Ordnance Systems Command
Washington, D.C. 20360

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DEPARTMENT OF THE NAVY
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-82524A(OS) TARGET POSITION SIMULATOR MK 26 MOD 0 (For Fire Control System MK 112)

ORGANIZATION *(Of submitter)*		CITY AND STATE
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8870

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